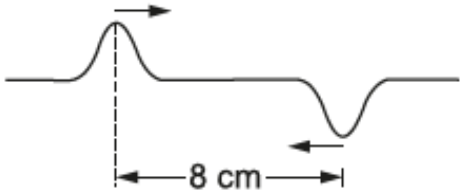
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two pulses in a stretched string whose centres are initially $8\ \text{cm}$ apart are moving towards each other, as shown in the figure. The speed of each pulse is $2\ \text{cm s}^{-1}$. After $2\ \text{s}$, the total energy of the pulses will be



(A) zero

(B) purely kinetic

(C) purely potential

(D) partly kinetic and partly potential

Q.2

A point mass is subjected to two simultaneous sinusoidal displacements in the x-directions: $x_1(t) = A\sin\omega t$ and $x_2(t) = A\sin(\omega t + 2\pi/3)$. Adding a third sinusoidal displacement $x(t) = B\sin(\omega t + \phi)$ brings the mass to a complete rest. The values of B and ϕ are respectively

(A) $\sqrt{2}A$ and $\frac{3\pi}{4}$

(B) A and $\frac{4\pi}{3}$

(C) $\sqrt{3}A$ and $\frac{5\pi}{6}$

(D) A and $\frac{\pi}{3}$

Q.3

The ends of a stretched wire of length L are fixed at $x = 0$ and $x = L$. In one experiment, the displacement of the wire is $y_1 = A\sin(\pi x/L)\sin\omega t$ and the energy is E_1, and in another experiment the displacement is $y_2 = A\sin(2\pi x/L)\sin 2\omega t$ and the energy is E_2. Then

(A) $E_2 = E_1$

(B) $E_2 = 2E_1$

(C) $E_2 = 4E_1$

(D) $E_2 = 16E_1$

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Q.4

A vibrating string of certain length ℓ under a tension T resonates with a mode corresponding to the first overtone (third harmonic) of an air column of length $75\ \mathrm{cm}$ inside a tube closed at one end. The string also generates 4 beats per second when excited along with a tuning fork of frequency n. Now when the tension of the string is slightly increased the number of beats reduces 2 per second. Assuming the velocity of sound in air to be $340\ \mathrm{m/s}$, the frequency n of the tuning fork in Hz is

(A) 344

(B) 336

(C) 117.3

(D) 109.3

Q.5

Two vibrating strings made of the same material but of lengths L and $2L$ have the radii $2r$ and r respectively. They are stretched under the same tension. Both the strings vibrate in their fundamental modes-the one of length L with frequency v_1 and the other with frequency v_2. The ratio v_1/v_2 is equal to

(A) 2

(B) 4

(C) 8

(D) 1

Q.6

A sonometer wire resonates with a given tuning fork, forming standing waves with five antinodes between the two bridges when a mass of $9\ \mathrm{kg}$ is suspended from the wire. When this mass is replaced by a mass M, the wire resonates with the same tuning fork, forming three antinodes for the same positions of the bridges. The value of M is

(A) $25\ \mathrm{kg}$

(B) $5\ \mathrm{kg}$

(C) $12.5\ \mathrm{kg}$

(D) $\frac{1}{25}\ \mathrm{kg}$

Q.7

A student is performing the experiment of resonance column. The diameter of the column tube is $4\ \mathrm{cm}$. The frequency of the tuning fork is $512\ \mathrm{Hz}$. The air temperature is $38^{\circ}\mathrm{C}$ in which the speed of sound is $336\ \mathrm{m/s}$. The zero of the meter scale coincides with the top end of the resonance column tube. When the first resonance occurs, the reading of the water level in the column is

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(A) $14.0\ \text{cm}$

(B) $15.2\ \text{cm}$

(C) $16.4\ \text{cm}$

(D) $17.6\ \text{cm}$

Q.8

A student is performing an experiment with a resonance column. The diameter of the column tube is $4\ \text{cm}$. The frequency of the tuning fork is $512\ \text{Hz}$. The air temperature is 38°C in which the speed of sound is $336\ \text{m s}^{-1}$. The zero of the metre scale coincides with the top end of the resonance column tube. When the first resonance occurs, the reading of the water level in the column is

(A) $14.0\ \text{cm}$

(B) $15.2\ \text{cm}$

(C) $16.4\ \text{cm}$

(D) $17.6\ \text{cm}$

Q.9

A hollow pipe of length $0.8\ \text{m}$ is closed at one end. At its open end, a 0.5-m-long uniform string is vibrating in its second harmonic and it resonates with the fundamental frequency of the pipe. If the tension in the wire is $50\ \text{N}$ and the speed of sound is $320\ \text{m s}^{-1}$, the mass of the string is

(A) 5 grams

(B) 10 grams

(C) 20 grams

(D) 40 grams

Q.10

A train moves towards a stationary observer with a speed of $34\ \text{ms}^{-1}$. The train sounds a whistle and its frequency registered by the observer is f_1. If the train's speed is reduced to $17\ \text{ms}^{-1}$, the frequency registered is f_2. If the speed of sound is $340\ \text{ms}^{-1}$ then the ratio f_1/f_2 is

(A) $\frac{18}{19}$

(B) $\frac{1}{2}$

(C) 2

(D) $\frac{19}{18}$

Q.11

A siren placed at a railway platform is emitting sound waves of 5kHz frequency. A passenger sitting in a

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moving train A records a frequency of $5.5\mathrm{kHz}$ while the train approaches the siren. During his return journey by a different train B, he records a frequency of $6.0\mathrm{kHz}$ while approaching the same siren. The ratio of the velocity of the train B to that of the train A is

(A) $242 : 252$

(B) $2 : 1$

(C) $5 : 1$

(D) $11 : 6$

Q.12

A police car with a siren of frequency of $8\mathrm{kHz}$ is moving with a uniform velocity of $36\ \mathrm{km\ h^{-1}}$ towards a tall building which reflects the sound waves. The speed of sound in air is $320\ \mathrm{m\ s^{-1}}$. The frequency of the siren heard by the car driver is

(A) $8.50\mathrm{kHz}$

(B) $8.25\mathrm{kHz}$

(C) $7.75\mathrm{kHz}$

(D) $7.50\mathrm{kHz}$

Q.13

Two monatomic ideal gases 1 and 2 of molecular masses m_1 and m_2 respectively are enclosed in separate containers kept at the same temperature. The ratio of the speed of sound in gas 1 to that in gas 2 is given by

(A) $\sqrt{\frac{m_1}{m_2}}$

(B) $\sqrt{\frac{m_2}{m_1}}$

(C) $\frac{m_1}{m_2}$

(D) $\frac{m_2}{m_1}$

Q.14

A vibrating string of length l under the tension T resonates with a mode corresponding to the first overtone (third harmonic) of an air column of length $75\ \mathrm{cm}$ inside a tube closed at one end. The string also generates 4 beats per second when excited with a tuning fork of frequency n. When the tension of the string is slightly increased, the beat frequency reduces to 2 beats per second. Assuming the speed of sound in air to be $340\ \mathrm{m\ s^{-1}}$, the frequency n of the tuning fork is

(A) $344\ \mathrm{Hz}$

(B) $336\ \mathrm{Hz}$

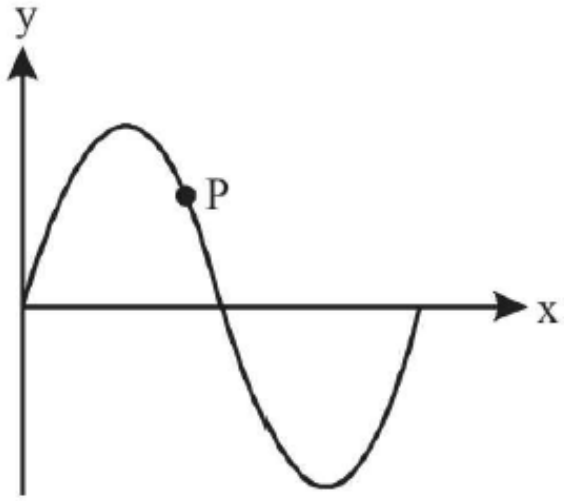
(C) $117.3\ \mathrm{Hz}$

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(D) $109.3\ \text{Hz}$

Q.15

A transverse sinusoidal wave moves along a string in the positive x-direction at a speed of $10\ \text{cm/s}$. The wavelength of the wave is $0.5\ \text{m}$ and its amplitude is $10\ \text{cm}$. Ata particular time t, the snap-shot of the wave is shown in figure. The velocity of point P when its displacement is $5\ \text{cm}$ is -



(A) $\frac{\sqrt{3}\pi}{50}\hat{\text{j}}\text{m/s}$

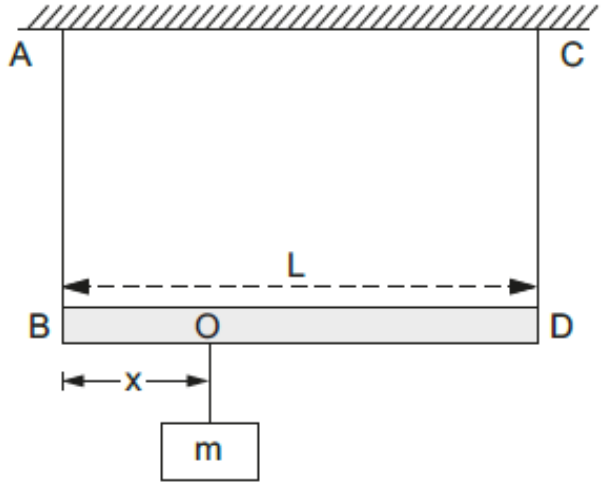
(B) $-\frac{\sqrt{3}\pi}{50}\hat{\text{j}}\text{m/s}$

(C) $\frac{\sqrt{3}\pi}{50}\hat{\text{i}}\text{m/s}$

(D) $-\frac{\sqrt{3}\pi}{50}\hat{\text{i}}\text{m/s}$

Q.16

A massless rod is suspended by two identical strings, AB and CD, of equal lengths. A block of mass m is suspended from the point C such that $\text{BO} = x$, as shown in the figure. Further, it is observed that the frequency of the first harmonic (fundamental frequency) in AB is equal to the 2nd harmonic in CD. Then the length of BO is



(A) $\frac{L}{5}$

(B) $\frac{4L}{5}$

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(C) $\frac{3L}{4}$

(D) $\frac{L}{4}$

Q.17

Two vibrating strings of the same material but lengths L and $2L$ have radii $2r$ and r respectively. They are stretched under the same tension. Both the strings vibrate in their fundamental nodes, the one of length L with frequency v_1 and the other with frequency v_2. The raio v_1/v_2 is given by

(A) 2

(B) 4

(C) 8

(D) 1

Q.18

In the experiment to determine the speed of sound using a resonance column,

(A) the prongs of the tuning fork are kept in a vertical plane

(B) the prongs of the tuning fork are kept in a horizontal plane

(C) in one of the two resonances observed, the length of the

(D) in one of the two resonances observed, the length of the resonating air column is close to half the wavelength of sound in air

Q.19

A train moves towards a stationary observer with speed 34 $\mathrm{m/s}$. The train sounds a whistle and its frequency registered by the observer is f_1. If the train's speed is reduced to 17 $\mathrm{m/s}$, the frequency registered is f_2. If the speed of sound is $340\ \mathrm{m/s}$, then the ratio f_1/f_2 is

(A) $18/19$

(B) $1/2$

(C) 2

(D) $19/18$

Q.20

The driver of a car approaching a vertical wall notices that the frequency of the horn of his car changes from $400\ \mathrm{Hz}$ to $450\ \mathrm{Hz}$ after being reflected from the wall. Assuming speed of sound to be $340\ \mathrm{m/s}$, the speed of approach of car towards the wall is

(A) $10\ \mathrm{m/s}$

(B) $20\ \mathrm{m/s}$

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(C) $30\ \mathrm{m/s}$

(D) $40\ \mathrm{m/s}$

Q.21

An α-particle and a proton are accelerated from rest by a potential difference of $100\ \mathrm{V}$. After this, their de Broglie wavelengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_\mathrm{p}}{\lambda_\alpha}$, to the nearest integer, is

Q.22

The equation of a transverse wave travelling on a rope is given by $y = 10\ \sin\pi\left(0.01x - 2t\right)$ where y and x are in cm and t is in seconds. The maximum transverse speed of a particle in the rope is about ____ $\mathrm{cm/s}$ (Mark your answer to the nearest integer)

Q.23

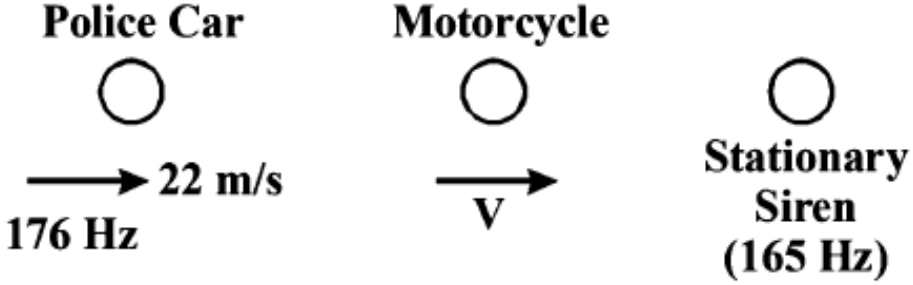
A silver sphere of radius $1\ \mathrm{cm}$ and work function $4.7\mathrm{eV}$ is suspended from an insulating thread in a free space. It is under continuous illumination of a 200-nm-wavelength light. As photoelectrons are emitted, the sphere gets charged and acquires a potential. The maximum number of photoelectrons emitted from the sphere is $A \times 10^n$ (where $1 < A < 10$). The value of n is

Q.24

The work function of silver and sodium are $4.6\mathrm{eV}$ and $2.3\mathrm{eV}$ respectively. The ratio of the slope of the stopping potential versus frequency plot for silver to that of sodium is

Q.25

A police car moving at $22\ \mathrm{m/s}$, chases a motorcyclist. The police man sounds his horn at $176\ \mathrm{Hz}$, while both of them move towards a stationary siren of frequency $165\ \mathrm{Hz}$. Calculate the speed of the motorcycle, if it is given that he does not observes any beats.



Q.26

When two progressive waves $y_1 = 4\sin(2x - 6t)$ and $y_2 = 3\sin\left(2x - 6t - \frac{\pi}{2}\right)$ are superimposed, the amplitude of the resultant wave is

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Q.27

A $20\ \text{cm}$ long string, having a mass of $1.0\ \text{g}$, is fixed at both the ends. The tension in the string is $0.5\ \text{N}$. The string is set into vibrations using an external vibrator of frequency $100\ \text{Hz}$. Find the separation (in cm) between the successive nodes on the string.

Q.28

A stationary source emits sound of frequency $f_0 = 492\ \text{Hz}$. The sound is reflected by a large car approaching the source with a speed of $2\ \text{m s}^{-1}$. The reflected signal is received by the source and superposed with the original. What will be the beat frequency of the resulting signal in Hz ? (Given that the speed of sound in air is $330\ \text{m s}^{-1}$ and the car reflects the sound at the frequency it has received.)

Q.29

Four harmonic waves of equal frequencies and equal intensities I_0 have phase angles $0, \frac{\pi}{3}, \frac{2\pi}{3}$ and π. When they are superposed, the intensity of the resulting wave is nI_0. The value of n is

Q.30

A stationary source is emitting sound at a fixed frequency f_0, which is reflected by two cars approaching the source. The difference between the frequencies of sound reflected from the cars is 1.2% of f_0. What is the difference in the speeds of the cars (in km per hour) to the nearest integer? The cars are moving at constant speeds much smaller than the speed of sound which is $330\ \text{ms}^{-1}$.

Answers & Solutions

Q.1 Answer:

purely kinetic

Solution:

After 2 s, the pulses will overlap completely. The string becomes straight and therefore does not have any potential energy. Its entire energy must be kinetic.

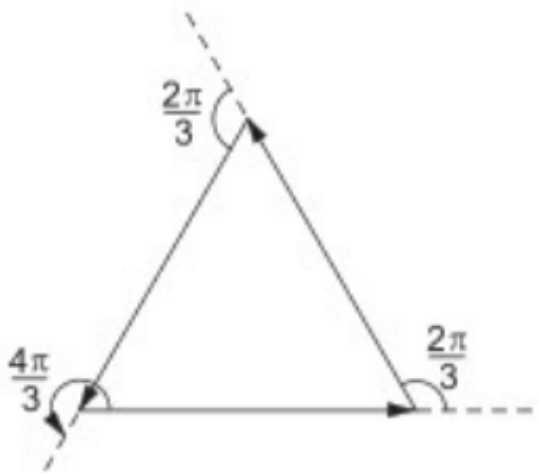
Q.2 Answer:

A and $\frac{4\pi}{3}$

Solution:

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Conclude from the vector triangle (equilateral).



Q.3 Answer:

$E_2 = 4E_1$

Solution:

A stationary wave has the equation of the form

$y = A \sin kx \sin \omega t$

Here, for y_1, we have $k_1 = \frac{\pi}{L}$ and $\omega_1 = \omega$

$\therefore \quad v_1 = \frac{\omega_1}{k_1} = \frac{\omega L}{\pi}$

For y_2, we have $k_2 = \frac{2\pi}{L}$ and $\omega_2 = 2\omega$

$\therefore \quad v_2 = \frac{\omega_2}{k_2} = \frac{\omega L}{\pi} = v_1$

Thus, the wave velocities are the same in both the cases. Also, they have the same amplitude. The frequency for y_2 is twice the frequency for y_1

Now, since energy $\propto$ (frequency) ^{2}

$\therefore \quad E_2 = 4E_1$

Q.4 Answer:

344

Solution:

The frequency (v) produced by the air column is given by

$\lambda \times v = \mathrm{v} \Rightarrow v = \frac{\mathrm{v}}{\lambda}$

Also, $\frac{3\lambda}{4} = \ell = 75 \text{ cm} = 0.75 \text{ m}$

Q.5 Answer:

1

Solution:

$v = \frac{1}{2l}\sqrt{\frac{T}{m}} = \frac{1}{2l}\sqrt{\frac{T}{A\rho}} = \frac{1}{2l}\sqrt{\frac{T}{\pi R^2 \rho}} = \frac{1}{2lR}\sqrt{\frac{T}{\pi\rho}}$

As T and ρ are the same for both the wires, $v \propto \frac{1}{lR}$. Here, $LR = L \times 2r = 2L \times r.$ $\quad \therefore v_1 =$

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v_2

Q.6 Answer:

25 kg

Solution:

The frequency of vibration of a string is given by

$n = \frac{N}{2l}\sqrt{\frac{T}{m}}$

where N = number of loops (or segments) = number of antinodes. The other symbols have their usual meanings.

Here, $n = \frac{5}{2l}\sqrt{\frac{9g}{m}} = \frac{3}{2l}\sqrt{\frac{Mg}{m}}$ **or** $M = 25 \text{ kg}.$

Q.7 Answer:

15.2 cm

Solution:

Considering the end correction $[e = 0.3D]$**, we get**

$$L + e = \frac{\lambda}{4} \quad \Rightarrow L = \frac{\lambda}{4} - e$$
$$= \frac{336 \times 100}{512 \times 4} - 0.3 \times 4 = 15.2 \text{ cm} \left[\because \lambda = \frac{v}{v}\right]$$

Q.8 Answer:

15.2 cm

Solution:

$\frac{\lambda}{4} = L + e$
$\Rightarrow L = \frac{\lambda}{4} - e = \frac{v}{4f} - 0.6R = 16.4 \text{ cm} - 1.2 \text{ cm} = 15.2 \text{ cm}$

Q.9 Answer:

10 grams

Solution:

$\frac{V_{\text{sound}}}{2l_{\text{pipe}}} = 2\left(\frac{1}{2l_{\text{string}}}\right)\sqrt{\frac{T}{m}}$
mass of the string $= \text{m}l_{\text{s}} = 10 \text{ g}$

Q.10 Answer:

$\frac{19}{18}$

Solution:

$n_{\text{o}} = \frac{n_{\text{s}}V}{V - v_{\text{s}}}$
Here, $f_1 = \frac{n_{\text{s}}V}{340 \text{ m s}^{-1} - 34 \text{ m s}^{-1}}$ **and** $f_2 = \frac{n_{\text{s}}V}{340 \text{ m s}^{-1} - 17 \text{ m s}^{-1}}$

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$\therefore \quad \frac{f_1}{f_2} = \frac{340\ \text{m s}^{-1} - 17\ \text{m s}^{-1}}{340\ \text{m s}^{-1} - 34\ \text{m s}^{-1}} = \frac{19}{18}$

Q.11 Answer:

$2 : 1$

Solution:

When an observer approaches a stationary source, which is emitting sound waves of frequency n_s, he hears a frequency n, given by

$n = n_\text{s}\left(1 + \frac{v}{V}\right)$

where v = velocity of the observer and V = velocity of sound. Here, let v_A and v_B be the velocities of the trains A and B.

For A, we have $n = 5.5\text{kHz} = (5\text{kHz})\left(1 + \frac{v_\text{A}}{V}\right)$ **or** $\frac{v_\text{A}}{V} = 0.1$

For B, we have $n = 6\text{kHz} = (5\text{kHz})\left(1 + \frac{v_\text{B}}{V}\right)$ **or** $\frac{v_\text{B}}{V} = 0.2$

$\therefore \quad \frac{v_\text{B}}{v_\text{A}} = 2 \quad$ **or** $\quad v_\text{B} : v_\text{A} = 2 : 1.$

Q.12 Answer:

8.50kHz

Solution:

$f' = \left(\frac{v + v_0}{v - v_\text{s}}\right) f = \left(\frac{320 + 10}{320 - 10}\right)\left(8 \times 10^3\ \text{Hz}\right) = 8.5\text{kHz}$

Q.13 Answer:

$\sqrt{\frac{m_2}{m_1}}$

Solution:

KEY CONCEPT : $C_\text{ms} = \sqrt{\left(\frac{\gamma RT}{M}\right)}$ **Here** $C_\text{mms} \propto \sqrt{\frac{1}{m}};$

$\therefore \frac{C_{\text{ms}_1}}{C_{\text{rms}_2}} = \sqrt{\left(\frac{m_2}{m_1}\right)}$

Q.14 Answer:

$344\ \text{Hz}$

Solution:

The frequency of the third harmonic of the closed pipe is

$f = 3\left(\frac{v}{4l}\right) = \frac{3 \times 340\ \text{m s}^{-1}}{4 \times 0.75\ \text{m}} = 340\ \text{Hz}$

Since the increase in tension reduces the beat frequency from $4\ \text{s}^{-1}$ to $2\ \text{s}^{-1}$,

$\therefore n - 340\ \text{Hz} = 4\ \text{Hz} \quad \Rightarrow \quad n = 344\ \text{Hz}$

Q.15 Answer:

$\frac{\sqrt{3}\pi}{50}\hat{\text{j}}\text{m/s}$

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Solution:

Since the wave is sinusoidal moving in positive x-axis the point will move parallel to y-axis therefore options (c) and (d) are ruled out. As the wave moves forward in positive X-direction, the point should move upwards i.e. in the positive Y-direction. Therefore correct option is a.

Q.16 Answer:

$\frac{L}{5}$

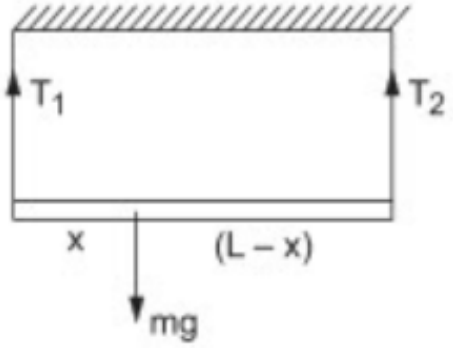
Solution:

$T_1 + T_2 = mg, T_1 x = T_2(L - x)$

$f_1 = \frac{1}{2l}\sqrt{\frac{T_1}{\mu}}$ and $f_2 = \frac{2}{2l}\sqrt{\frac{T_2}{\mu}}$

where μ = linear mass density and $f_1 = f_2$

$\therefore \quad T_1 = 4T_2 \quad$ **or** $\quad T_1 x = \frac{T_1}{4}(L - x) \quad$ **or** $\quad x = \frac{L}{5}$



Q.17 Answer:

1

Solution:

$(n_1 = \frac{1}{2\ell}\sqrt{\left(\frac{T}{4\pi r^2 \rho}\right)}$ **and** $n_2 = \frac{1}{4\ell}\sqrt{\left(\frac{T}{\pi r^2 \rho}\right)}$

$n = \frac{v}{\lambda} = \frac{1}{\lambda}\sqrt{\frac{T}{m}}$ **[where** $\frac{\lambda}{2}$ **= length of string]**

$\therefore \frac{n_1}{n_2} = 2 \times \frac{1}{2} = 1 \left[\because m = \frac{\text{mass}}{\text{length}} = \frac{\rho \times A \times \text{length}}{\text{length}} = \rho A\right])$

Q.18 Answer:

the prongs of the tuning fork are kept in a vertical plane

Solution:

The prongs of the tuning fork are kept in a vertical plane in order to set up longitudinal standing waves in the air column.

Q.19 Answer:

19/18

Solution:

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$n_1 = n_0 \frac{340}{340-34} = \frac{10}{9} n_0$
$n_2 = n_0 \frac{340}{340-17} = \frac{20}{19} n_0; \quad \frac{n_1}{n_2} = \frac{10}{9} \times \frac{19}{20} = \frac{19}{18}$

Q.20 Answer:

20 m/s

Solution:

$450 = 400 \left(\frac{340+v_s}{340-v_s} \right)$
$\Rightarrow \frac{9}{8} = \frac{340+v_s}{340-v_s}$
$\Rightarrow \quad 9(340) - 9v_s = 8(340) + 8v_s$
$\Rightarrow \quad 17v_s = 340$
$\Rightarrow v_s = 20 \text{ m/s}$

Q.21 Answer:

—

Solution:

Kinetic energy $= \mathrm{K} \cdot \mathrm{E} = \mathrm{qV}$
De Broglie wavelength $= \frac{\mathrm{h}}{\sqrt{2\mathrm{mK.E}}} = \frac{\mathrm{h}}{\sqrt{2\mathrm{mqV}}}$
Therefore ratio $\frac{\lambda_\mathrm{p}}{\lambda_\alpha} = \frac{\sqrt{\mathrm{m}_\alpha \mathrm{q}_\alpha}}{\sqrt{\mathrm{m}_\mathrm{p} \mathrm{q}_\mathrm{p}}}$
Alpha particle has 4 times the mass and twice the charge of a proton.
$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{8} \approx 3$

Q.22 Answer:

—

Solution:

The given equation is $y = 10\sin(0.01\pi x - 2\pi t)$
Comparing with the general wave equation $y = A\sin kx - \omega t$
Hence, $\omega =$ **coefficient of** $t = 2\pi$
Amplitude, $A = 10$
$\Rightarrow$ **Maximum speed of the particle,** $v_{\max} = A\omega$
$\Rightarrow \quad v_{\max} = 10 \times 2\pi$
$\Rightarrow \quad v_{\max} = 10 \times 2 \times 3.14$
$\Rightarrow \quad v_{\max} = 62.8 \approx 63 \quad \text{cm} \quad \text{s}^{-1}$

Q.23 Answer:

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Solution:

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Here, radius of sphere $R = 1 \text{ cm} = 1 \times 10^{-2} \text{ m}$

Work function, $W = 4.7\text{eV}$

Energy of incident radiation

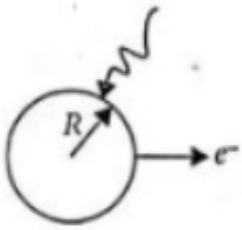
$= \frac{hc}{\lambda} = \frac{1240\text{eVnm}}{200 \text{ nm}} = 6.2\text{eV}$

According to Einsteins photoelectric equation

$\frac{hc}{\lambda} = \phi + eV_s$

$6.2\text{eV} = 4.7\text{eV} + eV_s$

$V_s = 1.5 \text{ V}$



The sphere will stop emitting photoelectrons, when the potential on its surface becomes equal to 1.5 V

$\therefore \frac{1}{4\pi\varepsilon_0}\frac{Q}{R} = 1.5$

$\Rightarrow \frac{1}{4\pi\varepsilon_0}\frac{Ne}{R} = 1.5$

where $N =$ **Number of photoelectrons emitted**

$e =$ **charge of each electron**

$N = \frac{1.5 \times R}{\frac{1}{4\pi\tau_0} \times e} = \frac{1.5 \times 1 \times 10^{-2}}{9 \times 10^9 \times 1.6 \times 10^{-19}}$

$N = \frac{15}{16} \times \frac{1}{9} \times 10^8 = \frac{5}{48} \times 10^8$

$N = \frac{50}{48} \times 10^7 = 1.04 \times 10^7$

$\therefore Z = 7$

Q.24 Answer:

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Solution:

For photoelectric effect: $\frac{h\nu}{e} - \frac{\phi}{e} = V_0$

The slope is: $\tan\theta = \frac{h}{e} =$ **constant**

$\therefore$ **The ratio will be 1.**

Q.25 Answer:

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Solution:

$f_1 =$ **frequency of the police car heard by motorcyclist,** $f_2 =$ **frequency of the siren heard by motorcyclist.**

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$f_1 = \frac{330-v}{330-22} \times 176; f_2 = \frac{330+v}{330} \times 165$
$\because \quad f_1 - f_2 = 0 \Rightarrow v = 22 \text{ m/s}$

Q.26 Answer:

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Solution:

Resultant amplitude, $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$
$= \sqrt{4^2 + 3^2 + 2 \times 4 \times 3 \times \cos\frac{\pi}{2}} = \sqrt{16 + 9 + 0} = 5$

Q.27 Answer:

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Solution:

We know that, $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{0.5}{10^{-3}/0.2}} = 10 \text{ m/s}$ **The wavelength of the wave established**
$\lambda = \frac{v}{f} = \frac{10}{100} = 0.1 \text{ m} = 10 \text{ cm}$
The gap(separation) between two successive nodes(or antinodes $) = \lambda/2$
$= 10/2 \text{ cm}$
$= 5 \text{ cm}$

Q.28 Answer:

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Solution:

Frequency received by the approaching car (as observer)
$f_1 = \left(\frac{v+v_0}{v}\right) f$
Frequency of the reflected wave as received by the source
$f_2 = \left(\frac{v}{v-v_0}\right) f_1 = \left(\frac{v+v_0}{v-v_0}\right) f$
$\therefore$ beat frequency $= f_2 - f = \left(\frac{v+v_0}{v-v_0} - 1\right) f = \left(\frac{332}{328} - 1\right) 492 \text{ s}^{-1} = 6 \text{ s}^{-1}$

Q.29 Answer:

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Solution:

$y = \sqrt{I_0}\left[\sin O + \sin\frac{\pi}{3} + \sin\frac{2\pi}{3} + \sin\pi\right]$
$y = \sqrt{I_0}\left[\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\right] = \sqrt{3}\sqrt{I_0}$
$\therefore \quad I_r = y^2 = 3I_0 \quad \Rightarrow \quad n = 3$

Q.30 Answer:

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Solution:

Let v be the speed of sound and v_c and f_0 the speed and frequency of car.

The frequency of sound reflected by the car is

$$\therefore f_1' = f_0\left[\frac{v+v_c}{v-v_c}\right]$$

Differentiating the above equation w.r.t. v_c, we get

$$\frac{df_1'}{dv_c} = f_0\left[\frac{(v-v_c)\frac{d}{dv_c}(v+v_c)-(v+v_c)\frac{d}{dv_c}(v-v_c)}{(v-v_c)^2}\right]$$

$$\therefore \frac{df_1'}{dv_c} = f_0\left[\frac{2v}{(v-v_c)^2}\right] = f_0\frac{2v}{v^2}\ (\because v_c \ll< v)$$

$$\therefore \frac{df_1'}{f_0}\times 100 = \frac{2}{v}\times dv_c$$

$$\therefore 0.012 = \frac{2\times dv_c}{330} \qquad \therefore dv_c = 0.198\ \text{m/s} \approx 7\ \text{km/h}$$

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